

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.3_k1)

Metric

$$ds^2 = -Y(x_3, x_4)^2 \sin(x_1)^2 dx_2 \otimes dx_2 + Y(x_3, x_4)^2 dx_1 \otimes dx_1 + e^{(2\lambda(x_3, x_4))} dx_3 \otimes dx_3 + e^{(2\nu(x_3, x_4))} dx_4 \otimes dx_4$$

Coordinates

$$(x^\mu) = (x_2, x_1, x_3, x_4)$$

Metric Functions

$$\Sigma = \sin(x_1)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} Y(x_3, x_4) \sin(x_1) \right) dx_2 + \left(-\frac{1}{2} \sqrt{2} Y(x_3, x_4) \right) dx_1$$

$$n = \left(\frac{1}{2} \sqrt{2} Y(x_3, x_4) \sin(x_1) \right) dx_2 + \left(\frac{1}{2} \sqrt{2} Y(x_3, x_4) \right) dx_1$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{(\lambda(x_3, x_4))} \right) dx_3 + \left(\frac{1}{2} i \sqrt{2} e^{(\nu(x_3, x_4))} \right) dx_4$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{(\lambda(x_3, x_4))} \right) dx_3 + \left(-\frac{1}{2} i \sqrt{2} e^{(\nu(x_3, x_4))} \right) dx_4$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2Y(x_3, x_4) \sin(x_1)} \right) dx_2 + \left(-\frac{\sqrt{2}}{2Y(x_3, x_4)} \right) dx_1$$

$$n = \left(-\frac{\sqrt{2}}{2Y(x_3, x_4) \sin(x_1)} \right) dx_2 + \left(\frac{\sqrt{2}}{2Y(x_3, x_4)} \right) dx_1$$

$$m = \left(\frac{1}{2} \sqrt{2} e^{(-\lambda(x_3, x_4))} \right) dx_3 + \left(\frac{1}{2} i \sqrt{2} e^{(-\nu(x_3, x_4))} \right) dx_4$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} e^{(-\lambda(x_3, x_4))} \right) dx_3 + \left(-\frac{1}{2} i \sqrt{2} e^{(-\nu(x_3, x_4))} \right) dx_4$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2Y(x_3, x_4) \sin(x_1)} \partial_{x_2} - \frac{\sqrt{2}}{2Y(x_3, x_4)} \partial_{x_1}$$

$$\Delta = -\frac{\sqrt{2}}{2Y(x_3, x_4) \sin(x_1)} \partial_{x_2} + \frac{\sqrt{2}}{2Y(x_3, x_4)} \partial_{x_1}$$

$$\delta = \frac{1}{2} \sqrt{2} e^{(-\lambda(x_3, x_4))} \partial_{x_3} + \frac{1}{2} i \sqrt{2} e^{(-\nu(x_3, x_4))} \partial_{x_4}$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} e^{(-\lambda(x_3, x_4))} \partial_{x_3} - \frac{1}{2} i \sqrt{2} e^{(-\nu(x_3, x_4))} \partial_{x_4}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{1}{4} i \sqrt{2} e^{(-\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4) - \frac{1}{4} \sqrt{2} e^{(-\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \nu(x_3, x_4)$$

$$\beta = \frac{1}{4} i \sqrt{2} e^{(-\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4) + \frac{1}{4} \sqrt{2} e^{(-\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \nu(x_3, x_4)$$

$$\tau = -\frac{\sqrt{2} e^{(-\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4)}{2 Y(x_3, x_4)} - \frac{i \sqrt{2} e^{(-\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4)}{2 Y(x_3, x_4)}$$

$$\pi = \frac{\sqrt{2} e^{(-\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4)}{2 Y(x_3, x_4)} - \frac{i \sqrt{2} e^{(-\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4)}{2 Y(x_3, x_4)}$$

$$\epsilon = -\frac{\sqrt{2} \cos(x_1)}{4 Y(x_3, x_4) \sin(x_1)}$$

$$\gamma = -\frac{\sqrt{2} \cos(x_1)}{4 Y(x_3, x_4) \sin(x_1)}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\begin{aligned} \Phi_{02} = & \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \lambda(x_3, x_4)}{2Y(x_3, x_4)} + \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{2Y(x_3, x_4)} \\ & + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{2Y(x_3, x_4)} + \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4)}{2Y(x_3, x_4)} \\ & - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} Y(x_3, x_4)}{2Y(x_3, x_4)} - \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial^2}{\partial x_3 \partial x_4} Y(x_3, x_4)}{Y(x_3, x_4)} + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} Y(x_3, x_4)}{2Y(x_3, x_4)} \end{aligned}$$

$$\Phi_{10} = 0$$

$$\begin{aligned} \Phi_{11} = & -\frac{1}{4} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4)^2 + \frac{1}{4} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \lambda(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4) - \frac{1}{4} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \nu(x_3, x_4)^2 + \frac{1}{4} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4) \\ & - \frac{1}{4} e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} \lambda(x_3, x_4) - \frac{1}{4} e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} \nu(x_3, x_4) + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4)^2}{4Y(x_3, x_4)^2} + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4)^2}{4Y(x_3, x_4)^2} - \frac{1}{4Y(x_3, x_4)^2} \end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned} \Phi_{20} = & \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \lambda(x_3, x_4)}{2Y(x_3, x_4)} - \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{2Y(x_3, x_4)} \\ & + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{2Y(x_3, x_4)} - \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4)}{2Y(x_3, x_4)} \\ & - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} Y(x_3, x_4)}{2Y(x_3, x_4)} + \frac{i e^{(-\lambda(x_3, x_4) - \nu(x_3, x_4))} \frac{\partial^2}{\partial x_3 \partial x_4} Y(x_3, x_4)}{Y(x_3, x_4)} + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} Y(x_3, x_4)}{2Y(x_3, x_4)} \end{aligned}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\begin{aligned}
\Lambda = & -\frac{1}{12} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4)^2 + \frac{1}{12} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \lambda(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4) - \frac{1}{12} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \nu(x_3, x_4)^2 + \frac{1}{12} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4) \\
& + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \lambda(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{1}{12} e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} \lambda(x_3, x_4) \\
& - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{1}{12} e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} \nu(x_3, x_4) + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4)}{6 Y(x_3, x_4)} \\
& - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4)^2}{12 Y(x_3, x_4)^2} - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} Y(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4)^2}{12 Y(x_3, x_4)^2} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} Y(x_3, x_4)}{6 Y(x_3, x_4)} + \frac{1}{12 Y(x_3, x_4)^2}
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & \frac{1}{6} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4)^2 - \frac{1}{6} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \lambda(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4) + \frac{1}{6} e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} \nu(x_3, x_4)^2 - \frac{1}{6} e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} \lambda(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4) \\
& + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \lambda(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \lambda(x_3, x_4)}{6 Y(x_3, x_4)} + \frac{1}{6} e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} \lambda(x_3, x_4) \\
& - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4) \frac{\partial}{\partial x_3} \nu(x_3, x_4)}{6 Y(x_3, x_4)} + \frac{1}{6} e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} \nu(x_3, x_4) + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4) \frac{\partial}{\partial x_4} \nu(x_3, x_4)}{6 Y(x_3, x_4)} \\
& + \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial}{\partial x_3} Y(x_3, x_4)^2}{6 Y(x_3, x_4)^2} - \frac{e^{(-2\lambda(x_3, x_4))} \frac{\partial^2}{(\partial x_3)^2} Y(x_3, x_4)}{6 Y(x_3, x_4)} + \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial}{\partial x_4} Y(x_3, x_4)^2}{6 Y(x_3, x_4)^2} - \frac{e^{(-2\nu(x_3, x_4))} \frac{\partial^2}{(\partial x_4)^2} Y(x_3, x_4)}{6 Y(x_3, x_4)} - \frac{1}{6 Y(x_3, x_4)^2}
\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "